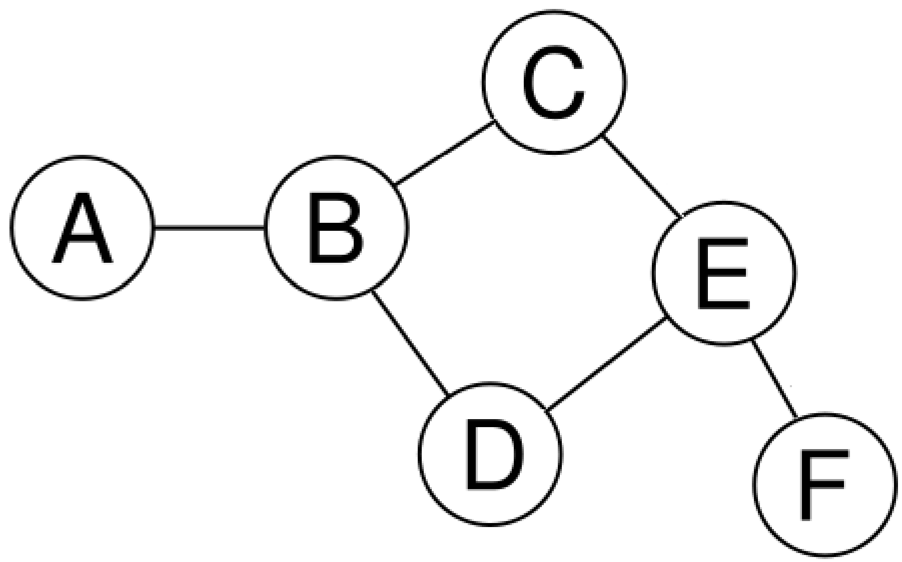
Graph Theory

A ***graph*** is a set of **nodes** (a.k.a. **vertices**) where some pairs are connected by **links** (a.k.a. **edges**).

```
V = {A, B, C, D, E, F}
E = {(A,B), (B,C), (B,D),
     (C,E), (D,E), (E,F)}
```

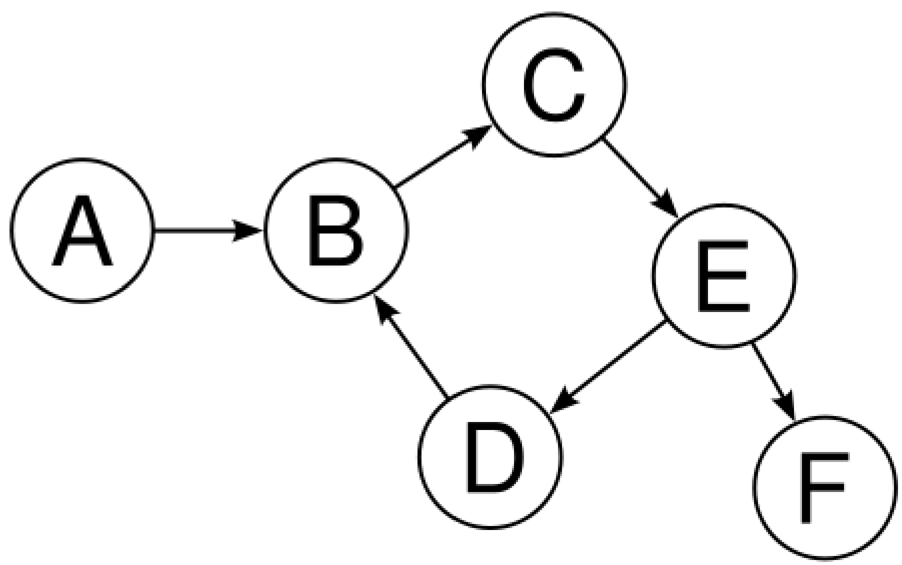
Undirected Graph

In an **undirected graph**, each edge links two vertices symmetrically. (i.e. every edge is bi-directional)



Directed Graph

In a **directed graph**, each edge links two vertices asymmetrically.



Edges list

The type of graph must be specified since the same lists of vertices is used for directed and undirected.

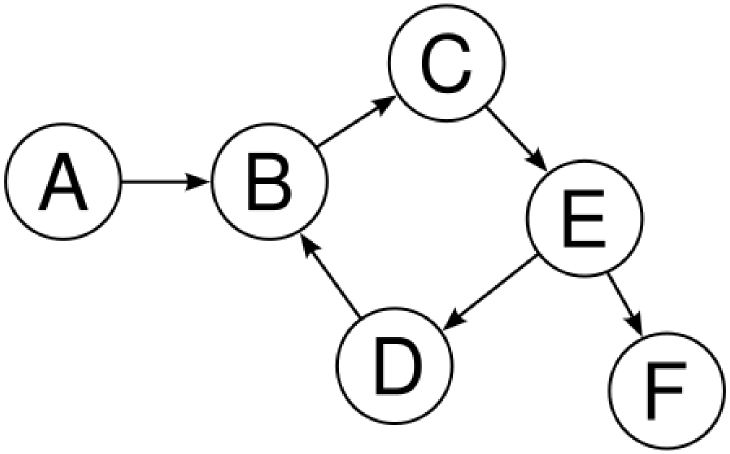
V = {A, B, C, D, E, F}

E = {(A,B), (B,C), (C,E),
(D,B), (E,D), (E,F)}

Adjacency Matrix for a Directed Graph

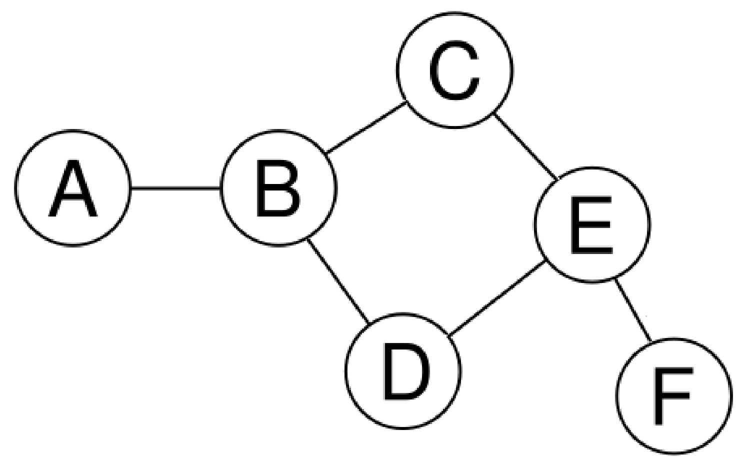
The row labels are the sources, and the column headers are the destinations.

	A	B	C	D	E	F
A	0	1	0	0	0	0
B	0	0	1	0	0	0
C	0	0	0	0	1	0
D	0	1	0	0	0	0
E	0	0	0	1	0	1
F	0	0	0	0	0	0



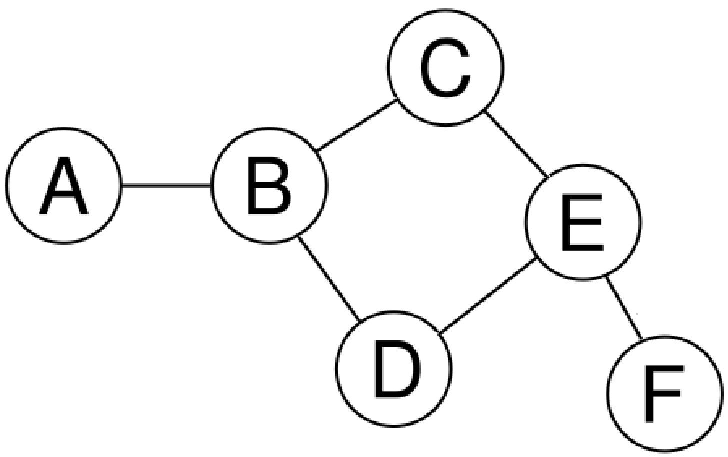
Adjacency Matrix for an Undirected Graph

	A	B	C	D	E	F
A	0	1	0	0	0	0
B	1	0	1	1	0	0
C	0	1	0	0	1	0
D	0	1	0	0	1	0
E	0	0	1	1	0	1
F	0	0	0	0	1	0



Adjacency Matrix for an Undirected Graph

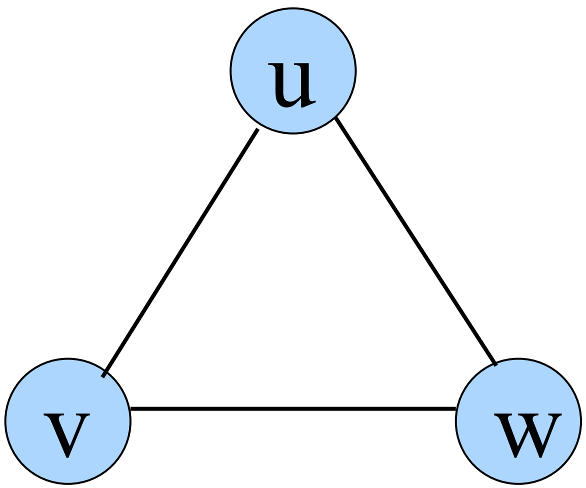
An undirected graph's matrix will be symmetric about the major diagonal.



	A	B	C	D	E	F
A	0	1	0	0	0	0
B	1	0	1	1	0	0
C	0	1	0	0	1	0
D	0	1	0	0	1	0
E	0	0	1	1	0	1
F	0	0	0	0	1	0

Adjacency Matrix Representation

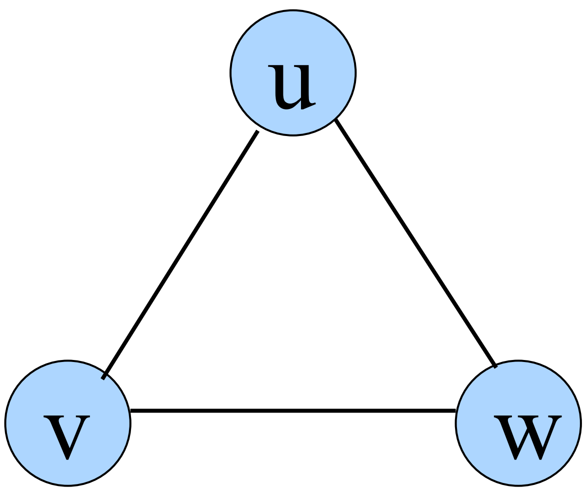
Example: Undirected Graph G (V, E)



	v	u	w
v	0	1	1
u	1	0	1
w	1	1	0

Adjacency List Representation

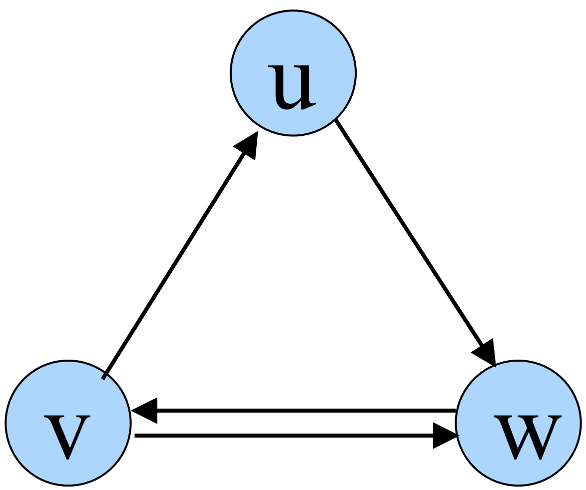
Each node (vertex) has a list of which nodes (vertex) it is adjacent.



node	Adjacency List
u	v , w
v	w, u
w	u , v

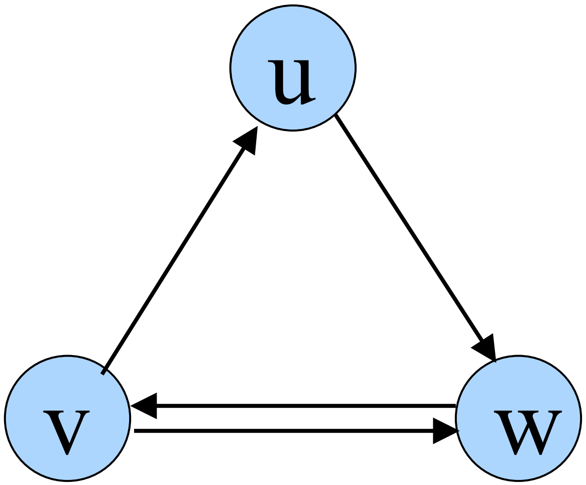
Adjacency Matrix Representation

Example: directed Graph G (V, E)



	v	u	w
v	0	1	1
u	0	0	1
w	1	0	0

Adjacency List Representation

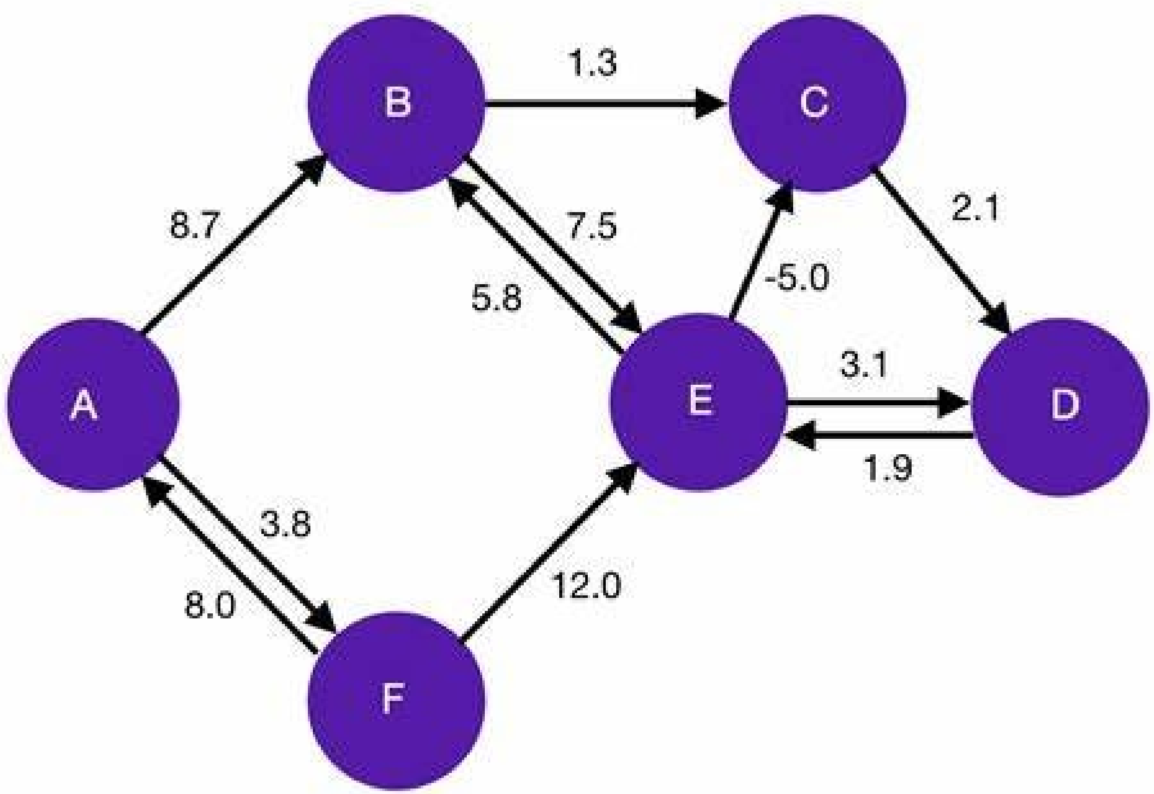


node	Adjacency List
u	w
v	u, w
w	v

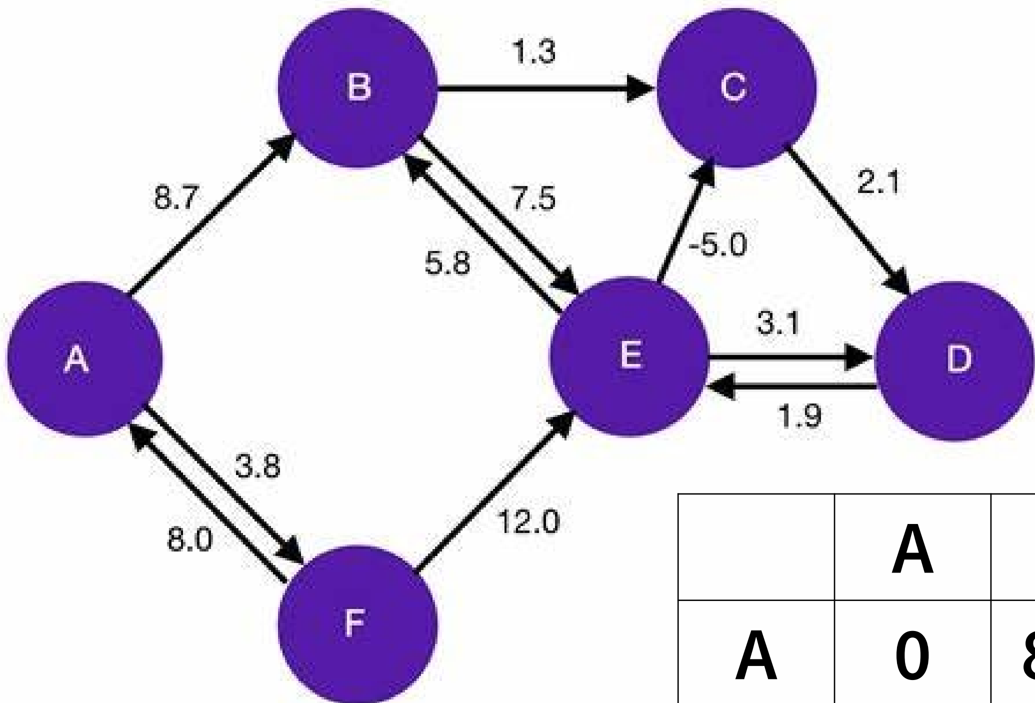
Note that here (v,w) and (w,v) are considered separate edges, while in an undirected graph they would be considered two ways to write the same edge <u>and wouldn't both be listed.</u>

Weighted Graph

A weighted graph associates a value (weight) with every edge in the graph.

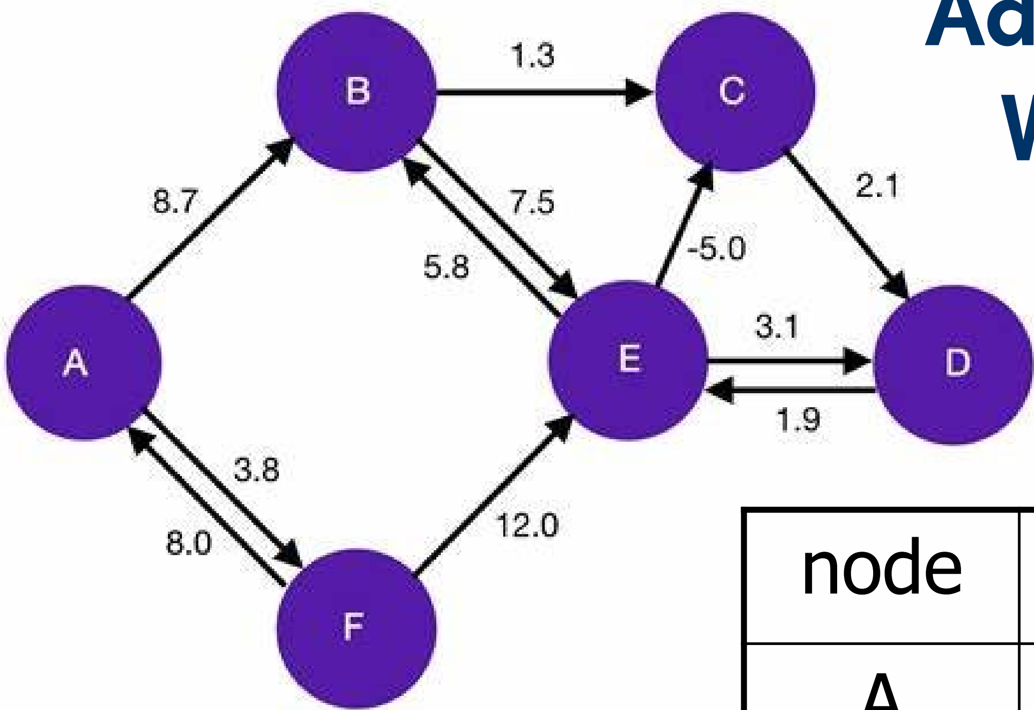


Adjacency Matrix for a Weighted Graph



	A	B	C	D	E	F
A	0	8.7	0	0	0	3.8
B	0	0	1.3	0	7.5	0
C	0	0	0	2.1	0	0
D	0	0	0	0	1.9	0
E	0	5.8	-5.0	3.1	0	0
F	8.0	0	0	0	12.0	0

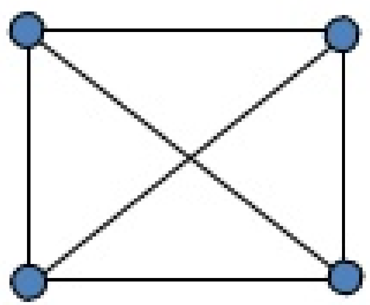
Adjacency List for a Weighted Graph



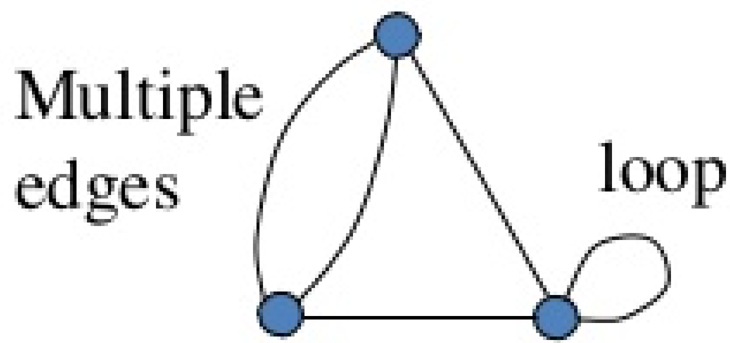
node	Adjacency List
A	B(8.7), F(3.8)
B	C(1.3), E(7.5)
C	D(2.1)
D	E(1.9)
E	B(5.8), C(-5.0), D(3.1)
F	A(8.0), E(12.0)

Simple vs Non-simple Graphs

A **simple graph** has no loops or multiple edges between nodes

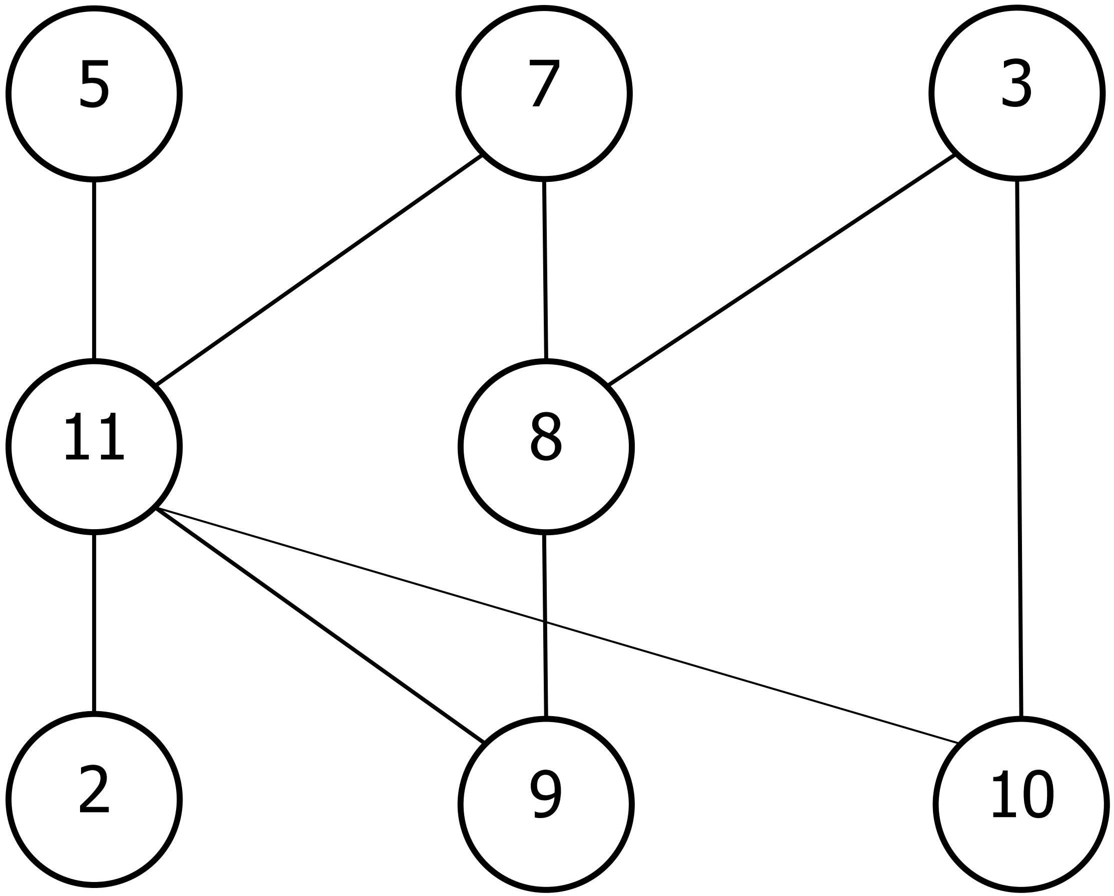


It is a simple graph.



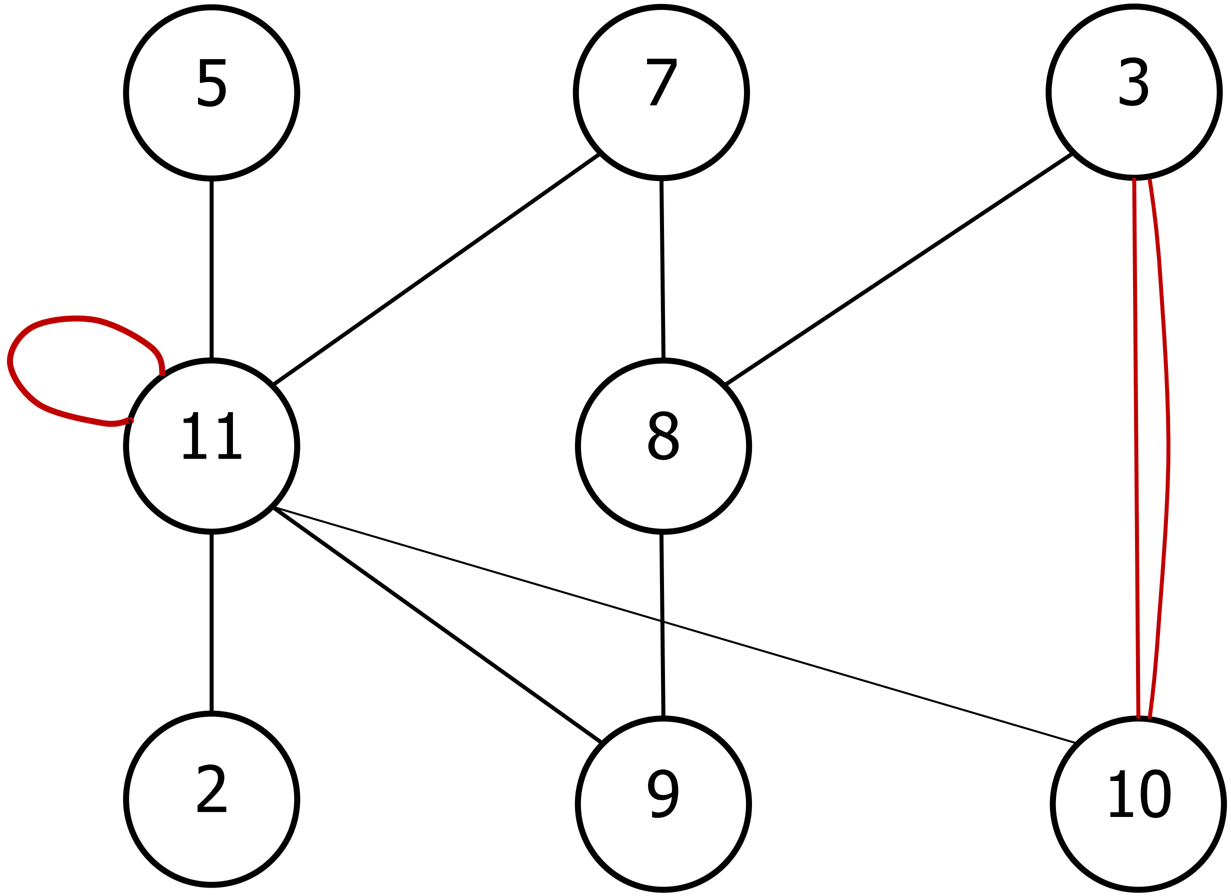
It is not simple.

A **simple undirected graph** never has multiple links between a pair of vertices. Nor does it have loops.



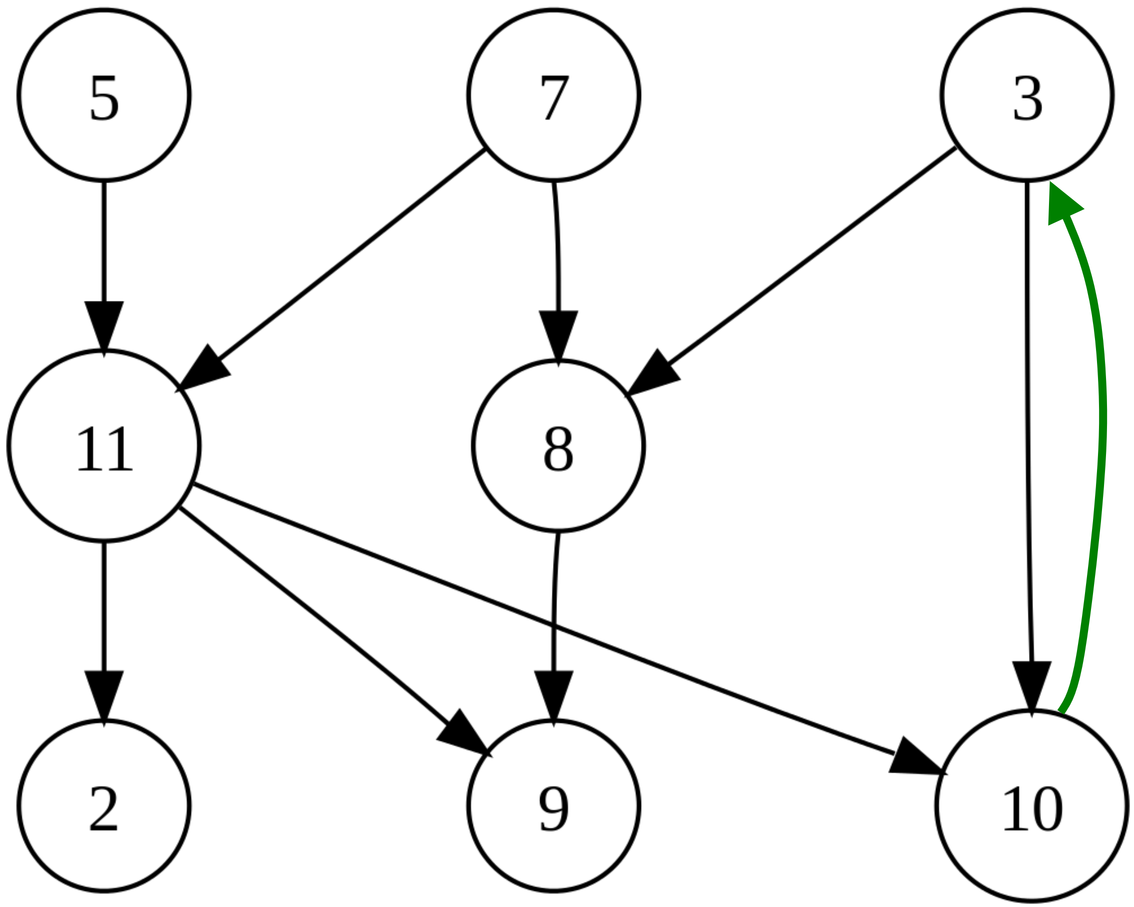
Note that this is an **unweighted graph.**

A **non-simple undirected graph** never has multiple links between a pair of vertices. Nor does it have loops.



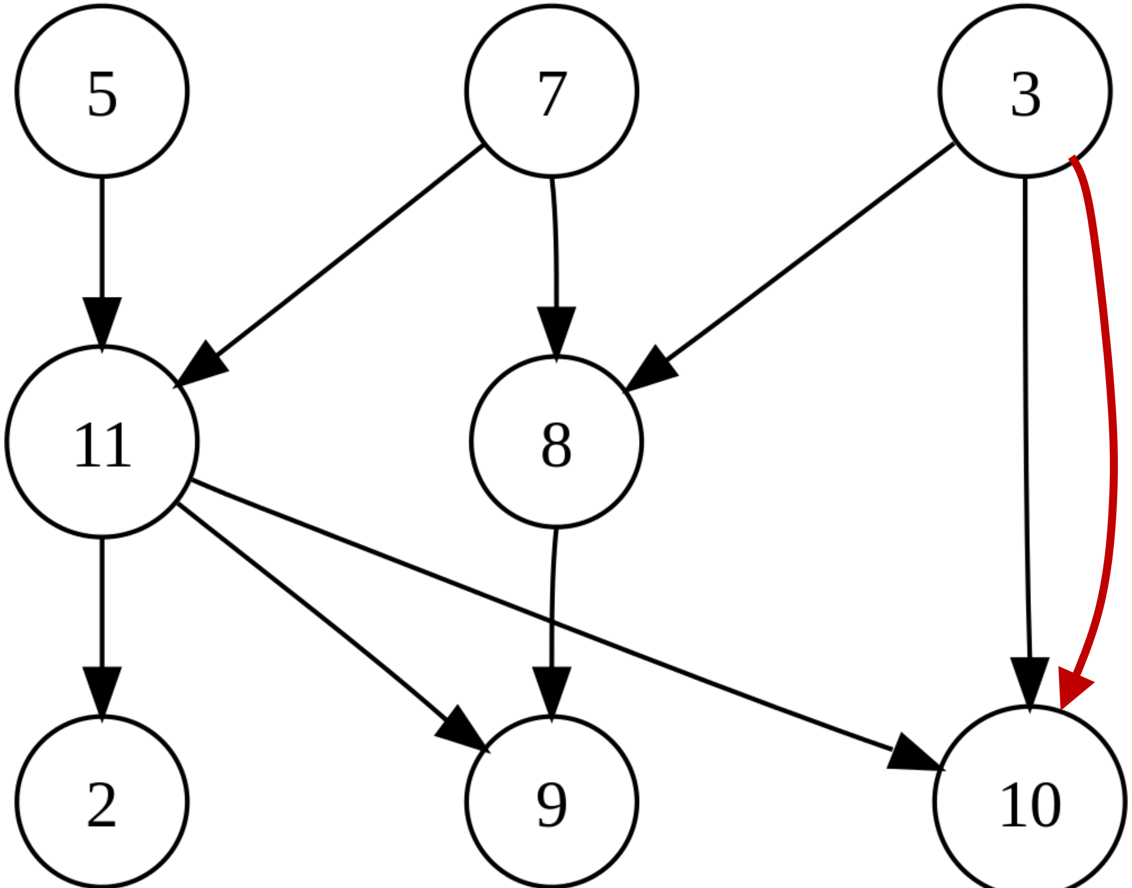
Note that this is an **unweighted graph.**

A **simple directed graph** has no multiple arrows with same source and target nodes.



Note that this is an **unweighted graph.**

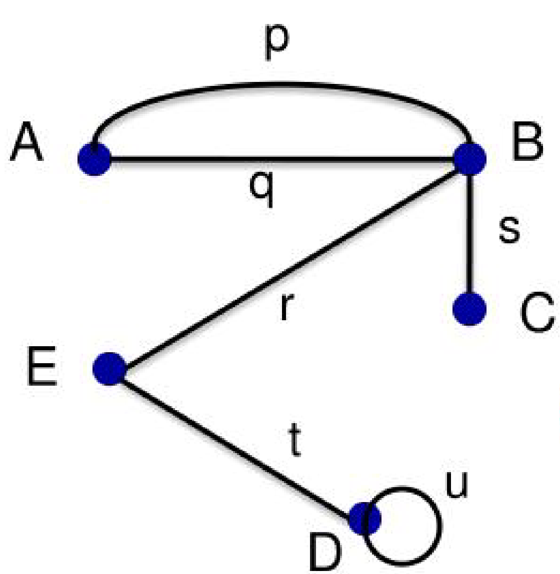
A **non-simple directed graph** has multiple arrows with same source and target nodes.



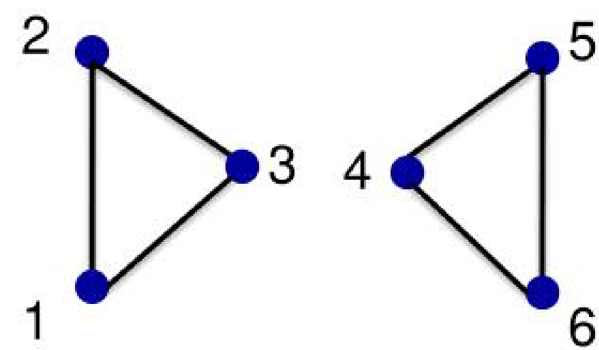
Note that this is an **unweighted graph.**

Connected/Disconnected Graphs & Components

A graph is connected if there is a path between any two vertices. In a disconnected graph the various connected pieces of a are called components of the graph.



Connected



Disconnected
Two components

Classify the graph:

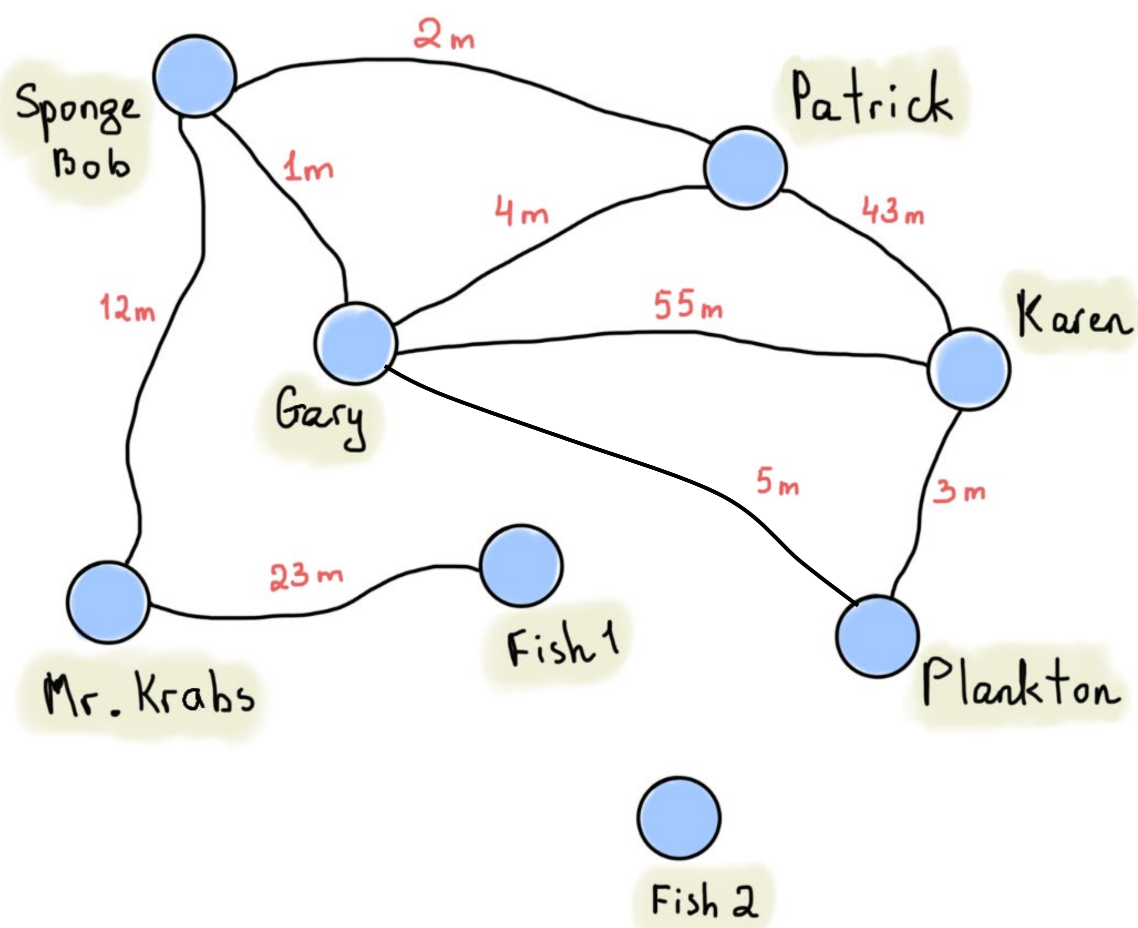
simple

weighted

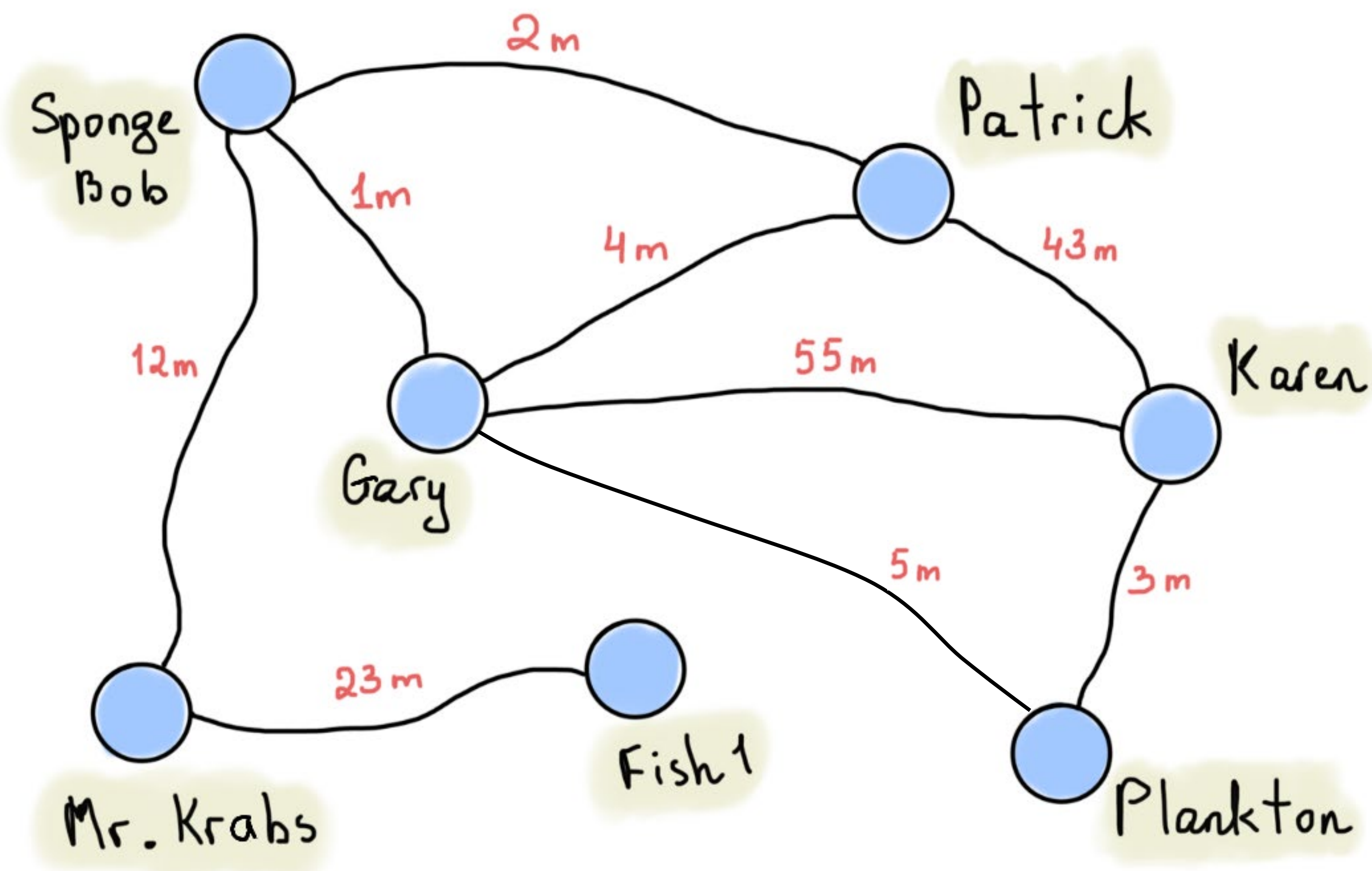
cyclic

undirected

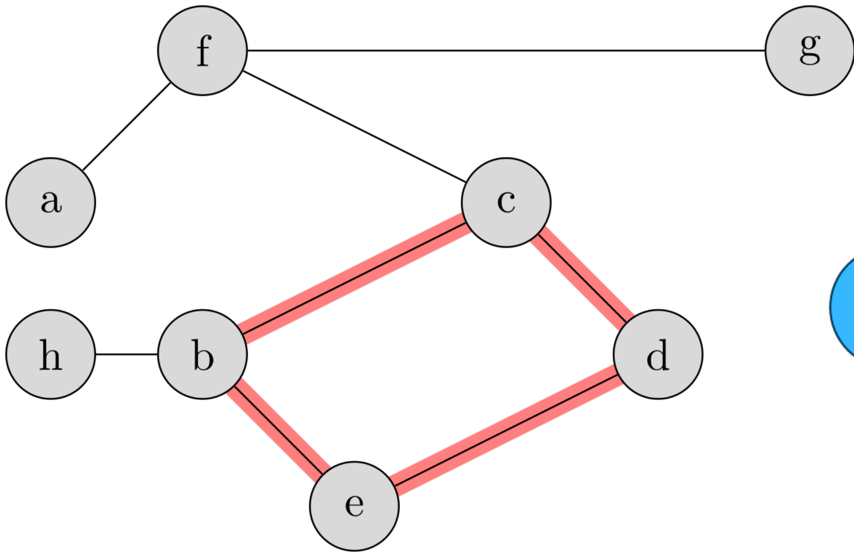
disconnected

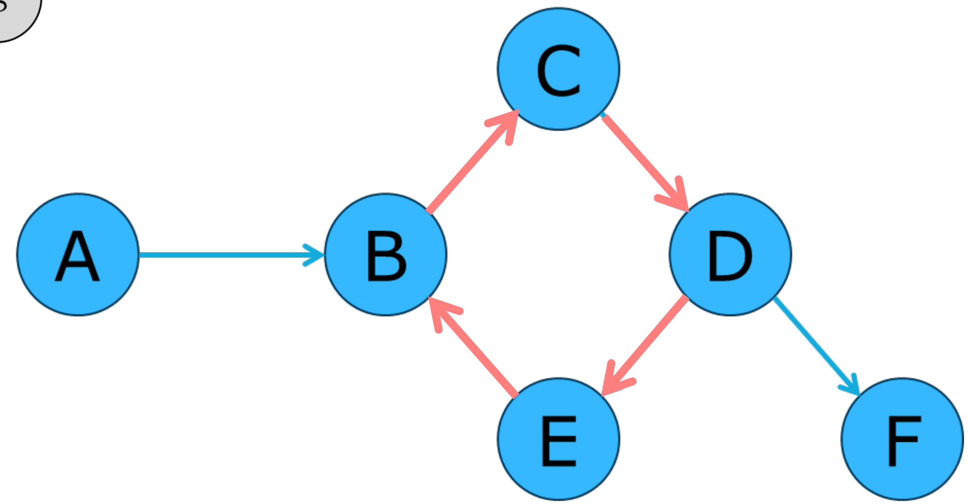


If Fish 2 was removed, it would be a simple connected graph, which is the type of graph we'll typically study.

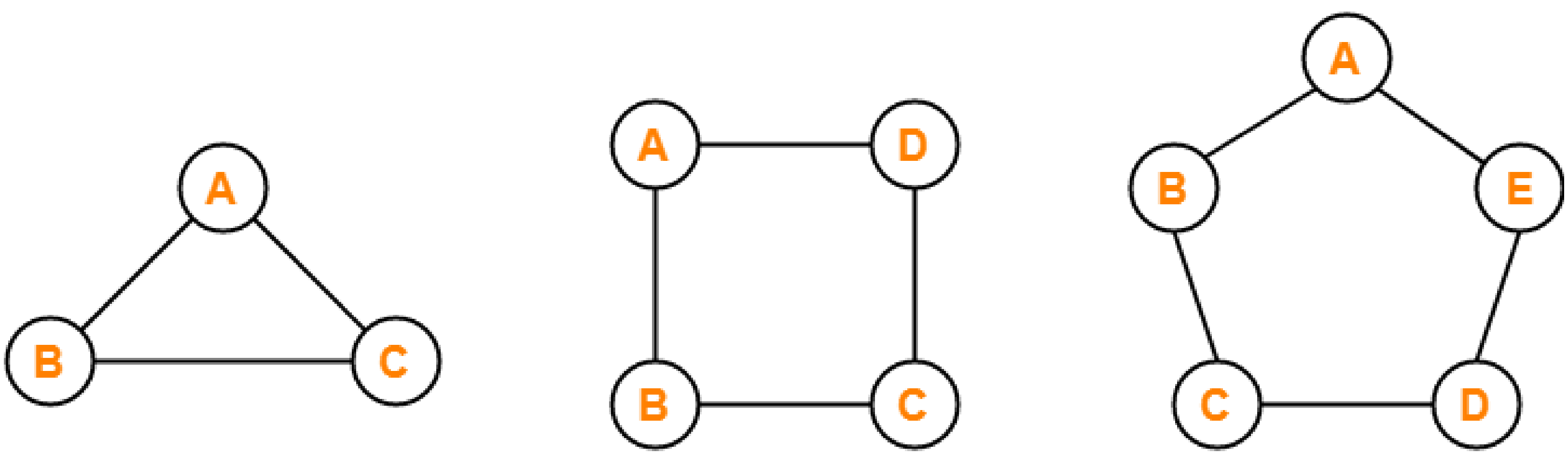


A **cyclic graph** <u>contains</u> a cycle.

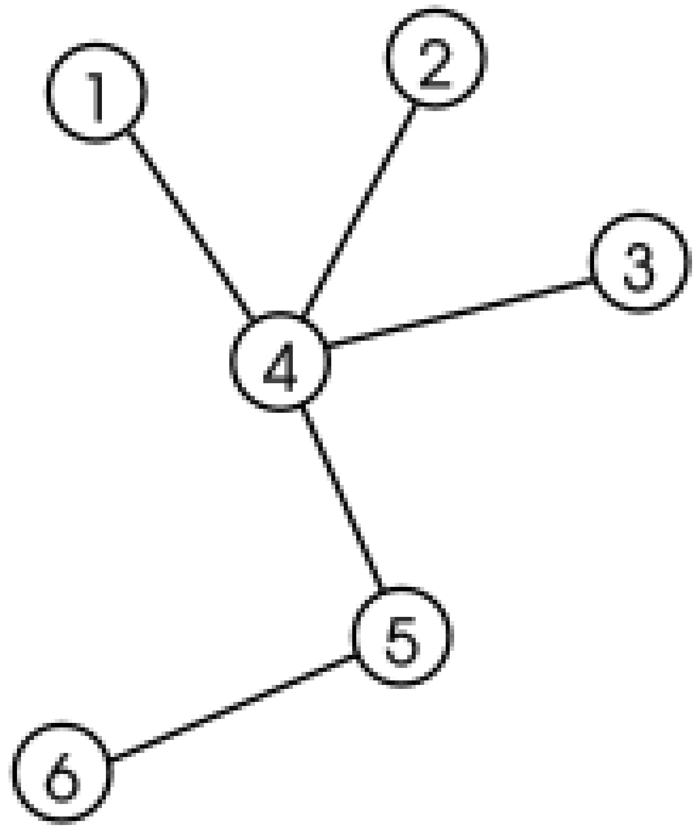




A **circular graph** <u>is</u> a cycle.

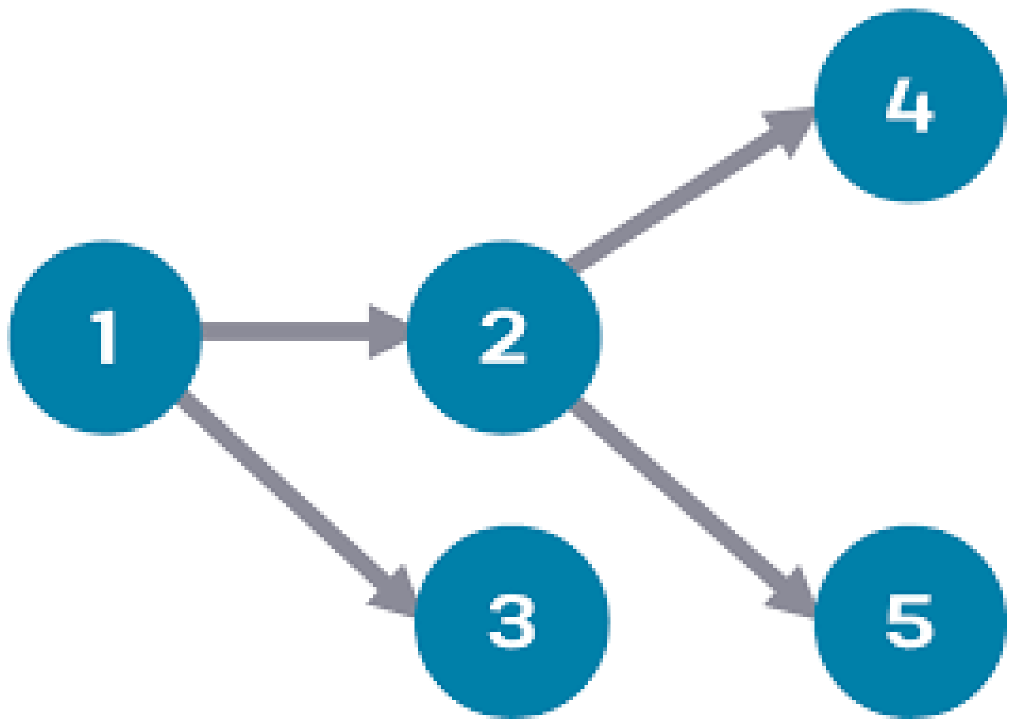


An **acyclic graph** contains no cycles.



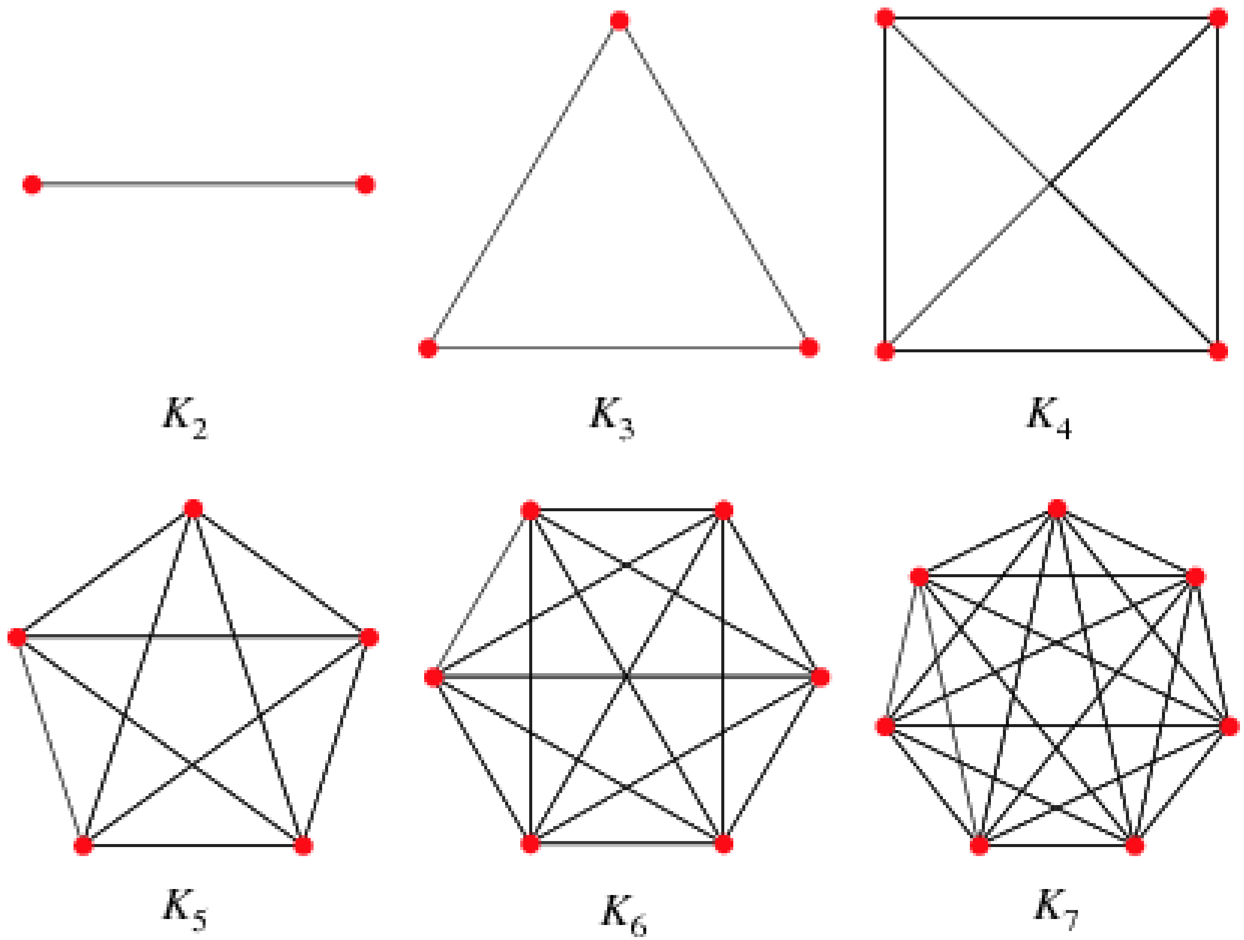
It's also known as a **tree**.

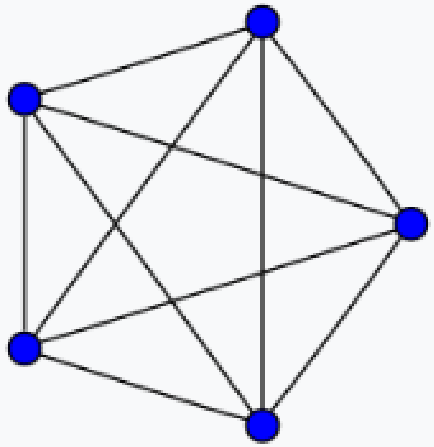
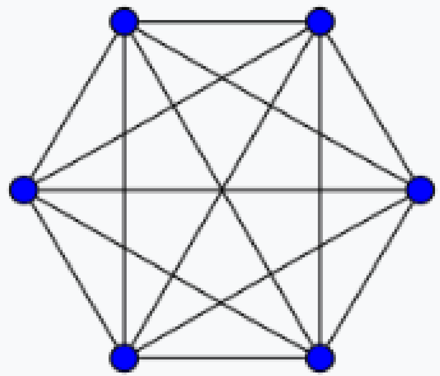
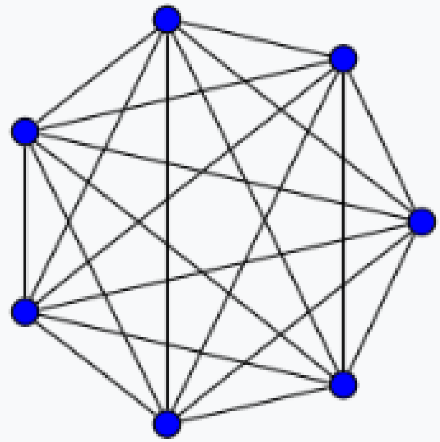
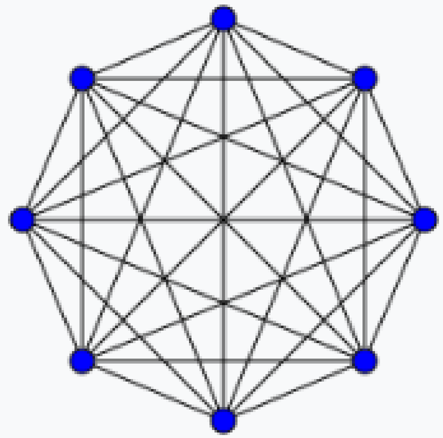
Directed acyclic graphs (DAGs) have numerous scientific and computational applications.



Complete Graph

A **simple undirected graph** in which every pair of distinct vertices is connected by a unique edge.



K_5: 10	K_6: 15	K_7: 21	K_8: 28
			
K_9: 36	K_{10}: 45	K_{11}: 55	K_{12}: 66